

**Isolation and identification of entomopathogenic fungi strains for Colorado potato beetle
(*Leptinotarsa decemlineata*) control**

Short title: **Biological control of potato pest**

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Abstract

Aims

The Colorado potato beetle, *Leptinotarsa decemlineata* (Say) (Coleoptera: Chrysomelidae) is the most widespread insect pest that causes major economic losses, especially on potatoes. Due to heavy insecticide use, this species now resists most pesticides, posing a significant control challenge. Frequent pesticide application also harms non-target organisms, the environment, and human health. Hence, utilizing biocontrol agents like entomopathogenic fungi (EPF) offers a viable alternative for pest management. The aim of this study was to identify and characterize new EPF strains isolated from soil samples and evaluate their efficacy against adult *L. decemlineata* under laboratory conditions.

Methods and Results

Soil samples were collected in potato fields or uncultivated areas adjacent to the field in the Czech Republic and EPF strains were isolated using a modified *Tenebrio* bait method. A total of 20 fungal strains were isolated and identified using morphological and molecular markers based on the 28S rRNA, ITS and elongation factor 1- α gene sequences as *Beauveria bassiana* (Bals.-Criv.) Vuill., *Beauveria brongniartii* (Sacc.) Petch and *Cordyceps fumosorosea* (Wize) Kepler, B. Shrestha & Spatafora (Hypocreales: Cordycipitaceae), *Purpureocillium lilacinum* (Thom.) Luangsa-ard, Houbraken, Hywel-Jones & Samson (Hypocreales: Ophiocordycipitaceae), *Metarhizium brunneum* (Petch) and *Metarhizium robertsii* Bisch., Rehner & Humber (Hypocreales: Clavicipitaceae). The bioassays revealed high variability among virulence of these strains against *L. decemlineata* with the shortest median time to death ($LT_{50} = 5.0$ days) in *M. robertsii* strain MAN3b.

Conclusions

Results shown that some EPF strains, particularly those of genera *Metarhizium*, can be promising biocontrol agents against the Colorado potato beetle.

Significance and Impact of the Study

EPF strains isolated in this study can be used for development of new mycoinsectides and applied in integrated pest management programme of potatoes.

Keywords: *Beauveria*, *Metarhizium*, *Cordyceps*, Chrysomelidae, biological pest control, virulence, genetic analysis

1. Introduction

The Colorado potato beetle *Leptinotarsa decemlineata* (Say) (CPB) is the most widespread and most voracious potato pest, causing great economic damage in the USA and Europe as well as in Asia (Hare 1990, Weber 2003, Alyokhin et al. 2022). The original distribution area of the CPB is North America. It first appeared on our continent in France in 1922, and by the end of the 20th century, it had become a problem throughout Europe, Asia Minor, Iran, Central Asia and Western China. In the Czech Republic, it was first discovered in 1945 in western and northern Bohemia, and by 1958 it had spread throughout the entire area, where it is still the largest potato pest today. In the early potato-growing areas of Polabí and in southern and south-eastern Moravia, it currently usually forms two generations (Doležal and Hausvater 2020).

The damage is caused by both larvae and adults, and the pest can also cause defoliation. The adults are characterized by high fecundity, one female can lay 300 to 800 eggs. The larva of the CPB consumes approximately 40 cm² of leaf area during its development, and an adult beetle requires a further 10 cm² per day. Yield losses can be up to 100% and therefore represent a limiting factor for potato production (Doležal and Hausvater 2018, Alyokhin et al. 2022).

The control of CPB has been based almost entirely on synthetic pesticides (Mota-Sanchez et al. 2006, Alyokhin et al. 2007). However, the massive use of insecticides to control CPB since the middle of the last century has not been able to prevent its spread, but on the contrary has significantly increased the resistance of CPB to most of the insecticides used, making it difficult to control at present (Zichová et al. 2010, Szendrei et al. 2012). Currently CPB has been found to be resistant to 55 different insecticides, including neonicotinoids and Spinosad, as well as the bacterium *Bacillus thuringiensis* (Kocourek et al. 2020). Moreover, many of these pesticides are toxic to the environment, having negative impacts on non-target organisms and leaving residues in food crops. For these reasons the range of approved insecticides against the CPB has changed considerably in recent years when usage of broad-spectrum pesticides such as organophosphates, carbamates and most neonicotinoid-based insecticides has been limited or completely banned by the European community (Robin and Marchard 2019). The number of newly approved insecticides to control pest infestations is limited and their use to ensure sufficient efficacy requires more frequent monitoring of pest infestations in the crop and more precise timing of treatments.

Therefore, various countries have undertaken efforts to develop and implement alternative, environmentally safe strategies for protecting crops against CPB. Particularly, the development of biological plant protection methods using natural enemies like entomopathogens seems to be a promising solution to the problem of pest resistance.

Entomopathogenic fungi (EPF) are leading among the potential alternative control agents of pest insects (Feron 1978, Vega et al. 2012, Quesada-Moraga 2020). Research has identified several fungal species as effective biological control agents against CPB. Many studies demonstrated high efficacy of *Beauveria bassiana* (Bals.-Criv.) Vuill. (Hypocreales: Cordycipitaceae) under laboratory (Ignoffo et al. 1983, Fargues et al. 1994, Todorova et al.

2000, Akbarian et al. 2012, Ropek and Kolodziejczyk 2019, Baki et al. 2021, Zemek et al. 2021, Keçili et al. 2022, Yıldırım et al. 2023) and field conditions (Roberts et al. 1981, Wraight and Ramos 2002, 2014). Research has also highlighted the potential of *Cordyceps fumosorosea* (Wize) Kepler, B. Shrestha & Spatafora (Hypocreales: Cordycipitaceae) (Hussein et al. 2016, Ropek and Kolodziejczyk 2019), *Cordyceps militaris* (L.) Fr. (Hypocreales: Cordycipitaceae) (Kryukov et al. 2014), *Metarhizium* spp. (Hypocreales: Clavicipitaceae) (Akbarian et al. 2012, Kryukov et al. 2017, Eski et al. 2022, Keçili et al. 2022) and *Purpureocillium lilacinum* (Thom) Luangsa-ard, Houbraken, Hywel-Jones & Samson (Hypocreales: Ophiocordycipitaceae) (Kepenekci et al. 2015) in controlling the CPB. The above mentioned studies documented large differences in efficacy among strains which, besides few commercial strains used, were isolated from soil (e.g. Keçili et al. 2022) or originated either from CPB (e.g. Zemek et al. 2021) or other insect hosts.

The aim of this study was to isolate novel strains of entomopathogenic fungi from soil samples collected from potato fields in the Czech Republic, and to identify and characterize these strains both morphologically and through molecular techniques. Additionally, the study aimed to evaluate the efficacy of these isolates against the wild Colorado potato beetle population under laboratory conditions. The wild population was used to ensure that the results are representative of natural, genetically diverse beetle populations, enhancing the applicability of the findings to real-world conditions. Bioassays were conducted using adult beetles because their role as the primary dispersers and reproducers makes them a critical target for control strategies. Additionally, CPB overwinters as adults in the soil, and larvae also pupate in the soil during the season. Therefore, targeting emerging adults with entomopathogenic fungi applied as a soil drench can be particularly effective (Půža et al. 2021). By focusing on adults, we can disrupt the life cycle at a crucial point, reducing the

population before they can reproduce and spread. Moreover, CPB adults are more resistant to EPF infection compared to its immature stages (Ignoffo et al. 1983, Fargues 1991, Long et al. 1998, Öztürk et al. 2015, Baki et al. 2021). Therefore, finding highly virulent strains against adults ensures high efficacy against all developmental stages of CPB.

2. Materials and Methods

2.1. Soil sampling

New strains of entomopathogenic fungi (EPF) were isolated from soil samples collected from commercial potato fields in South and West Bohemia, the Czech Republic, during the 2019 spring season (Table 1). Seven soil samples were collected from each site, five within the potato field (one from each corner and one from the center) and two from uncultivated boundaries next to the field. Each sample consisted of approximately 0.5 kg of soil taken with a shovel from a depth of 10-20 cm. The samples were placed individually in microtene bags, sealed, transported to the laboratory, stored in the dark at 20 ± 1 °C and processed within one week to minimize the decrease of conidia viability in soil samples (Correa et al. 2022).

2.2. Isolation of entomopathogenic fungi

Isolation of EPF was performed by a modified insect-bait method (Zimmermann 1986, Vänninen 1996). The yellow mealworm, *Tenebrio molitor* L. (Coleoptera: Tenebrionidae) larvae were used as bait-insect (Kim et al. 2018). Twenty grams of each soil sample was homogenized and placed to Petri dishes (vented, inner diameter 90 mm, height 15 mm, Gosselin™, Borre, France). The soil was moistened with water (1-2 mL) and ten larvae of *T. molitor*, were placed into each Petri dish. Then the Petri dishes were placed in a thermostat with a constant temperature of 25 ± 1 °C and constant darkness. After 14 days, grown EPF were isolated from the cadavers of *T. molitor* larvae showing mycosis symptoms. Isolation was

carried out on the artificial nutrient medium Potato Dextrose Agar (PDA) (Sigma-Aldrich, Darmstadt, Germany, 39 g/L) in the form of separation lines. After 7 days, the purity of the EPF culture was visually inspected for contamination by other fungi and/or bacteria. If contamination was detected, the EPF strains were purified by transferring them to new Petri dishes. The strains were purified using a selective medium based on PDA (Sigma-Aldrich, Darmstadt, Germany) with a fungicide dodine (0.05 g/L) and antibiotics cycloheximide (0.25 g/L) and chloramphenicol (0.5 g/L) (Sigma-Aldrich, Darmstadt, Germany) (Chase et al. 1986). Pure cultures of EPF strains were stored in the form of alginate pellets in a freezer at -22°C. The cultures are deposited at the Institute of Entomology, Biology Centre of the Czech Academy of Sciences, České Budějovice.

2.3. Fungal identification

The fungal species were determined using macroscopic and microscopic characters. The appearance on the infected *T. molitor* larvae and morphological characteristics were used as the preliminary identification of fungi. Morphological identification of fungal isolates was made by the colony morphology and determination of size and shape of, conidiogenous cells and conidia (Humber 1997, 2012) using an Olympus CH20 light microscope (Olympus Optical Co., Ltd., Tokyo, Japan), bright field, 400 × magnification. Morphological identification of the isolated fungi was also confirmed by gene sequencing and phylogenetic analysis.

2.4. Molecular analysis

2.4.1 DNA isolation, PCR amplification and DNA sequencing

DNA for genetic analysis was extracted from fresh mycelium grown at 25±1 °C for 10 days on Petri dishes with PDA medium. Each mycelium was collected in a sterile 1.5 mL

microtube. Genomic DNA was extracted from fungi using DEP-25 DNA Extraction Kit (Top-Bio Ltd., Vestec, Czech Republic) according to the manufacturer's recommendations.

The isolated strains were characterised with three molecular markers, part of the 28S rRNA (nrLSU), the internal transcribed spacer (ITS) and part of translation elongation factor 1-alpha (tef1), as suggested for fungal DNA barcoding (Raja et al. 2017). PCR amplifications were performed with optimisation according to Rehner and Buckley (2005). Reactions with a final volume of 12.5 µl consisted of 1 µl of isolated gDNA (not quantified), 0.75 µl of each primer (5 µM) and other components (10x green buffer, dNTPs, TaqUNIS DNA polymerase, PCR grade H₂O; Top-Bio, Ltd., Vestec, Czech Republic) according to the manufacturer's protocol. The PCR primers and cycle profiles are specified in Table 2. After performing the PCRs, all products were purified with an enzyme mixture (Exo I and FastAP; Thermo Fisher Scientific, Inc., Waltham, MA, USA) and sent to Eurofins Genomics, GmbH (Ebersberg, Germany) for sequencing. Chromatograms were manually processed and trimmed using Chromas Lite software, version 2.6.6. (Technelysium Pty, Ltd. South Brisbane, Australia); in the case of bidirectional sequencing, forward and reverse sequences were assembled into contigs in Lasergene v15 (DNASTAR, Inc., Madison, WI, USA). All sequences were deposited in GenBank.

2.4.2 Molecular identification of EPF species

The species were identified by comparing the DNA sequences in the NCBI GenBank using BLAST (Basic Local Alignment Search Tool; www.ncbi.nlm.nih.gov/BLAST/) searches. For phylogenetic placement, alignments of individual marker sequences were generated in MEGA version X (Kumar et al. 2018) using the MUSCLE algorithm and concatenated into a matrix of 2270 bp. Homologous sequences of reference species from GenBank (Table 3) were added and combined in the dataset. A phylogenetic tree was constructed from the combined dataset

with partitioning by marker in the IQ tree server (Nguyen et al. 2015), including both selection of the best substitution model (Model Finder, Kalyanamoorthy et al. 2017) and tree inference with the Maximum likelihood algorithm (ultra-fast bootstrap, 1000 replicates).

2.5 Preparation of fungal suspensions

The fungal strains were grown on Petri dishes (ø 90 mm) containing PDA. The plates were incubated at 25±1 °C in the dark for 10–14 days. The aerial conidia were harvested by scraping them into a sterile solution of 0.05% (v/v) Tween 80[®] (Sigma-Aldrich, Munich, Germany). The conidial suspension was filtered through sterile gauze to separate the mycelium and conidial clusters. In the uniform suspension, the spores were counted with a Neubauer improved counting chamber (Sigma-Aldrich, Munich, Germany) and, subsequently, the suspension was adjusted to a concentration of 1×10⁷ conidia per mL which is often used in insect bioassays. The suspension was left for approximately 12 h at a temperature 23±1 °C to accelerate and synchronize germination of conidia (Dillon and Charnley 1985, 1990) before its application. The viability of the spores was verified using a standard germination test (Skalický et al. 2014). Ten drops from suspension were applied using a 1 µL inoculation loop on the surface of 2% water agar, which was poured in a thin layer onto the surface of a sterile slide. After the drops had dried, the slides were moved into a wet chamber and incubated at 25±1 °C for 24 h. The percentage of germinating spores was determined using an Olympus CH20 light microscope (Olympus Optical Co., Ltd., Tokyo, Japan); bright field, 400× magnification. The spore germination of all strains was >98%.

2.6 Bioassays

The efficacy of individual fungal strains was tested on CPB adults collected in July and August 2020 from an organic potato field in Malonty, South Bohemia, Czech Republic

(48.7083920 N; 14.5778508 E). The beetles were individually immersed in the suspension of conidiospores of the fungus for 30 s (dip-test). All specimens of a control group were immersed in a sterile solution of 0.05% Tween® 80 only. The beetles were then placed into polystyrene Petri dishes lined with moist filter paper KA 0 (Papírna Perštein Ltd., Pernštejn, Czech Republic) and kept under constant conditions (25 ± 1 °C and 16L:8D photoperiod). The filter paper was daily moistened with distilled water to maintain optimal humidity in the Petri dishes and replaced with a new one once a week. The insects were monitored daily for four weeks to record insect mortality and the development of mycosis on the cadavers. Only mycosis verified by microscopic examination of the sporulating fungus was considered. The bioassays were repeated three times; each repetition was conducted in different period and tested new batch of twenty freshly-collected CPB adults per treatment. Conidia suspensions were prepared from new EPF cultures (PDA plates) for each repetition.

2.7 Statistical Analysis

The cumulative mortality data were performed in SAS® Studio (Release 3.81, Enterprise Edition, SAS Institute Inc., Cary, NC, USA) using the LIFETEST procedure of SAS/STAT® module (SAS Institute 2017). The Kaplan–Meier product limit estimate was used to determine both the mean and the median time to death (LT_{50} , the number of days until 50% of insects were dead) for each strain. The log-rank test statistics (PROC LIFETEST) was used to test the global hypothesis that mortality (time to death) differs between strains, and Šidák adjustment was used for multiple comparisons. Fungal infestation of cadavers at the end of the experiment was expressed as mean percentage of cadavers showing symptoms of mycosis. A generalized linear model with a binomial distribution and logit linkage was used to analyse the mycosis data. Treatment and replication were set as fixed effects. The analysis was performed using the GLM procedure (PROC GENMOD) of SAS/STAT® module. Means

were separated by the least-square means (LSMEANS) statement of SAS with Tukey-Kramer adjustment for multiple comparisons. *p* values <0.05 were considered statistically significant.

3 Results

Totally, 20 strains of EPFs were isolated from soil samples. They were identified as *B. bassiana*, *Beauveria brongniartii* (Sacc.) Petch, *C. fumosorosea*, *P. lilacinum*, *Metarhizium brunneum* (Petch) and *Metarhizium robertsii* Bisch., Rehner & Humber. More strains of entomopathogenic fungi were isolated from the uncultivated boundary (13 samples) than from the potato fields (7 samples) (Table 4).

3.1. Phylogenetic placement of the isolates

Phylogenetic analysis of all isolates (Fig. 1.) showed that with high support two isolates (BBA2b, BBA4b) clustered with *B. brongniartii*, three (BBA1b, BBA3b, BBA5b) with *B. bassiana*, two (IFR1b, IFR2b), with *C. fumosorosea*, one with *P. lilacinum* (IFR3b). The cluster of eight isolates (MAN1b, MAN2b, MAN4b, MAN5b, MAN6b, MAN7b, MAN8b, MAN9b) with *M. brunneum* also shows high support, while the support of the cluster of four isolates (MAN3b, MAN 10b, MAN11b, MAN12b) with *M. robertsii* is lower, but still quite acceptable. For all isolates, the species status was also confirmed morphologically.

3.2. Efficacy of EPF strains against CPB

The results of bioassays showed that all strains were pathogenic to CPB adults but large differences in the survival response of the beetles among strains were found (Fig. 2.). Survival analysis revealed a statistically highly significant effect of strain (log-rank test, $\chi^2 = 230.26$, $p < 0.0001$). The shortest median survival time ($LT_{50} = 5.0$ days) was estimated in beetles

treated with the MAN3b strain, while the longest ($LT_{50} = 17.0$ days) was found in the beetles treated with BBA2b strain, which did not significantly differ from that in control (Table 5).

As seen from Table 5, strains of *Metarhizium* sp. were the most virulent, while strains of other EPF species were less effective. Only one strain of *Beauveria* (BBA3b) and one strain of *Cordyceps* (IFR2b) turned out to affect significantly survival of treated beetles compared to control beetles.

Mycosis was observed in most cadavers at the end of experiment (Fig. 3.). Depending on the strain, mycosis ranged from 2.1 ± 2.1 to $78.3 \pm 8.8\%$ in IFR1b and MAN12b strains, respectively (Fig. 4.). The effect of treatment on mycosed cadavers was statistically highly significant ($\chi^2 = 247.34$, $df = 20$, $p < 0.0001$, while the effect of replication was not significant ($\chi^2 = 4.25$, $df = 1$, $p = 0.5143$). The pairwise comparison showed statistically significant differences among some strains (Fig. 4.).

Mycosis caused by *Beauveria* sp. were observed also in control group, where the average percentage of mycosed cadavers at the end of experiments was 16.1 ± 3.3 . This indicates that some adults in the field population used for bioassays were naturally infected with EPF. In addition to natural *Beauveria* infection, $5.8 \pm 2.4\%$ of beetles in the control were parasitized by nematodes belonging to the family Mermitidae.

4. Discussion

The Colorado potato beetle has developed resistance to all major classes of chemical insecticides and therefore alternative solutions are needed to provide safe and effective CPB control (Alyokhin et al. 2008, Chen et al. 2023). Fungal insect pathogens are important

natural control agents for many insects and other arthropod pests with a potential to significantly reduce host populations (Burgess 1981, Carruthers and Soper 1987, McCoy et al. 1988). Therefore, EPF are considered a promising tool for pest biocontrol globally (Skinner et al. 2014).

Several techniques for isolation of EPF from soil samples have been developed. In some studies EPF were isolated by culturing soil samples on selective media (Chase et al. 1986, Correa et al. 2022, Konopická et al. 2021, 2022). The most frequently used technique is based on the insect-bait method (Zimmermann 1986, Vänninen 1996). The isolation efficiency using this method was reported to be higher than the method of selective medium (Kim et al. 2018, Correa et al. 2022). Larva of the greater wax moth, *Galleria mellonella* L. (Lepidoptera: Pyralidae) is the most commonly used bait-insect but *T. molitor* larva is considered another insect model for isolation of EPF since it is easy to rear at a low cost (Kim et al. 2018). It was found, however, that recoveries of different EPF species from soil samples are affected by insect bait used (Sharma et al. 2018).

In this study, we isolated EPF from soil by 'Tenebrio-bait method' because we wanted to recover EPF virulent to beetle pest. We selected and identified 20 strains of EPF at species level using the macromorphological and micromorphological characteristics and molecular methods as: *B. bassiana*, *B. brongniartii*, *C. fumosorosea*, *P. lilacinum*, *M. brunneum* and *M. robertsii*. Phylogenetic analysis is necessary for the proper species identification as morphological characteristics are not always sufficient to distinguish some species of EPF. Because multilocus-based phylogenies are important for the accurate assignment of isolates to specific clades (Wang et al. 2020) we characterized the isolated strains with three molecular markers, a part of the 28S rRNA (nrLSU), the internal transcribed spacer (ITS) and a part of translation elongation factor 1-alpha (tef1), as recommended for fungal DNA barcoding (Raja

et al. 2017). Sequences of the barcoding markers were successfully obtained for all strains.

The consequent phylogenetic analysis provided unequivocal species identification which was corroborated by the morphological method.

The aim of the present study was to find highly virulent EPF strains which could be recommended for CPB biocontrol. Many previous laboratory and field studies (e.g. Wraight and Ramos 2002, 2014, Hussein et al. 2016, Yulin et al. 2018, Půža et al. 2021, Zemek et al. 2021) investigated potential of various EPF strains as biocontrol agents against *L. decemlineata*. Our study focused on indigenous Czech strains and their efficacy against a wild population of CPB.

The obtained results showed that all 20 strains were pathogenic against CPB adults in laboratory bioassays nevertheless a statistically highly significant differences in virulence among the strains were found. The estimated median lethal time (LT_{50}) ranged from 5 to 17 days, with the shortest LT_{50} found in species of EPF belonging to the genus *Metarhizium*. Strains of *Metarhizium* sp. turned out in this study to be the most virulent while strains of other EPF species were less effective. The most effective strain against CPB beetles was *M. robertsii* MAN3b when LT_{50} was only 5 days after treatment. In study by Keçili et al. (2022), various EPF were isolated from soil samples in potato fields. The identified species were tested against adults and larvae of CPB under laboratory conditions. The highest mortalities in adults population were obtained when CPB was treated by *B. bassiana* and *Metarhizium* sp. Among these, nine isolates of *B. bassiana* and *Metarhizium* sp. caused mortalities ranging from 96 to 83% within 15 days and they were significantly different from the control. The lower efficiency of the same species against larvae of CPB was described in the study by Kryukov et al. (2021). The author tested *M. robertsii* against larvae of CPB and 10 days after treatment, the topical infection with fungus caused 56.7% mortality in population. Larvae that died after treatment with the fungus were mummified (86%). On the contrary, the two *M.*

anisopliae isolates that Akbarian et al. (2012) tested against CPB larvae in their study were ineffective. DEMI001 and IRAN437C isolates of *M. anisopliae* showed the lowest mortality rate in among other isolates on second and fourth instars larvae. There were no significant differences in comparison with control treatment. Results of these experiments indicated that the fungal isolates caused a different mortality in CPB beetles or larvae.

Purpureocillium lilacinum strain IFR3b was found to be less effective against adults of CPB in our study ($LT_{50} = 14.0$ days). A study by Kepenekci et al. (2015) obtained similar results when testing the *P. lilacinum* against the last larval instar of CPB. Mortality was only 33.2% on the 10th day of treatment with a fungal concentration of 10^8 CFU mL⁻¹.

In our study, *Beauveria* sp. strains showed LT_{50} values ranging from 11.0 (BBA3b) to 17.0 (BBA2b) days. Only two out of five strains of *Beauveria* (BBA1b and BBA3b) significantly affected beetles survival compared to control treatment. The lower efficacy of *Beauveria* sp. may be associated with lower susceptibility of CPB adults to fungal infection. The differences in susceptibility to EPF among developmental stages of CPB have been documented for example by Fargues (1972, 1991) and Long et al. (1998) who reported much lower susceptibility of adults and eggs compared to larval or pupal stages. This was recently confirmed by Baki et al. (2021) who showed that some of the 14 Turkish *B. bassiana* isolates applied at a concentration of 1×10^7 conidia mL⁻¹ caused mortality of CPB larvae between 96.7 and 100% while they caused lower mortality in eggs and adults. In another study (Yıldırım et al. 2023), *B. bassiana* isolates applied at the same concentration against CPB larvae and adults caused corrected mortality between 10 and 80%, and between 3 and 50%, respectively. In contrast, a previous study by Zemek et al. (2021) reported up to 100% mortality in CPB adults treated by new native strains of *B. bassiana* applied at a concentration of 1×10^7 spores mL⁻¹ with LT_{50} for about 7 days only. The efficacy of these native strains was found to be significantly higher than that of the reference strain GH4 re-isolated from

commercial mycopesticide BotaniGard® WP (Certis USA, L.L.C., Butte, MT, USA). A study by Todorova et al. (2000) obtained similar results when tested ten isolates of *B. bassiana* for adult CPB at the same concentration. Six isolates were highly virulent and mortality ranged from 86 to 100% 6 days after treatment. All these results indicate that high between-strains variability in virulence against *L. decemlineata* adults exists and that native strains isolated from the same host might be more effective in CPB control.

Results obtained in the present study further revealed that *B. bassiana*-infected adults were observed also in a control group where the average percentage of mycosed cadavers at the end of experiments was 16.1%. This indicates that some proportion of adults used in bioassays which were collected from organic farm in Malonty was naturally infected by *Beauveria* sp.. Natural infection of CPB by *Beauveria* sp. was found also earlier (Zemek et al. 2021) when 8.3 % of beetles were naturally infected. This indicates that *Beauveria* sp. infection can be frequently found in field populations of CPB in the Czech Republic, at least in fields where no pesticides are applied. Mietkiewski et al. (1996) also found *B. bassiana* to be the dominant fungal pathogen infecting about 21% of diapausing beetles in Poland. In addition to natural *Beauveria* sp. infection, 5.8% of beetles in control were parasitized by nematodes belonging to the family Mermitidae. Mermithid nematodes have the potential to suppress populations of agricultural pest insects including grasshoppers, several beetles and caterpillars, and some leafhoppers, ants and other insect groups (Poinar 1988).

Another species of EPF tested against CPB was *C. fumosorosea*. The efficacy against CPB adults was lower compared to the strains of *Metarhizium* sp. in this study. Other studies have, however, reached different conclusions regarding the efficacy of *C. fumosorosea* against CPB. For example, Hussein et al. (2016) reported a high efficacy of strain CCM 8367 isolated from a lepidopteran pest (Zemek et al. 2007) against last instar CPB larvae. The highest larval mortality, reaching 93%, was observed on the 7th day after treating insects with blastospores

at a concentration of 5×10^7 spores mL⁻¹. This indicates that CPB larvae were as or more receptive to this fungal strain compared to larvae or pupae of Lepidoptera (Hussein et al. 2013, Prenerová et al. 2013, Zemek et al. 2020). These results confirm a relatively wide host range of this fungus (Zimmerman, 2008). Field trials of another study revealed that potato plots treated with *C. fumosorosea* or *B. bassiana* had significantly higher tuber yield compared to untreated control (Ropek and Kołodziejczyk 2019).

In conclusion, twenty new soil-born strains of entomopathogenic fungi were isolated. They belong to *B. bassiana*, *B. brongniartii*, *C. fumosorosea*, *P. lilacinum*, *M. brunneum* and *M. robertsii*. Species status was verified by both morphological as well as molecular methods, in which three DNA barcoding makers (nrLSU, ITS and tef1), were applied. The bioassays revealed high variability among virulence of these strains against *L. decemlineata* and indicate that particularly the species of the genus *Metarhizium* can be promising biocontrol agents against this important pest. Further research on field efficacy and mass production parameters of selected strains will be necessary before their commercial application.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability

The data that support the findings of this study are available upon request.

Ethical Statement

Not applicable.

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Author contributions

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Figure legends

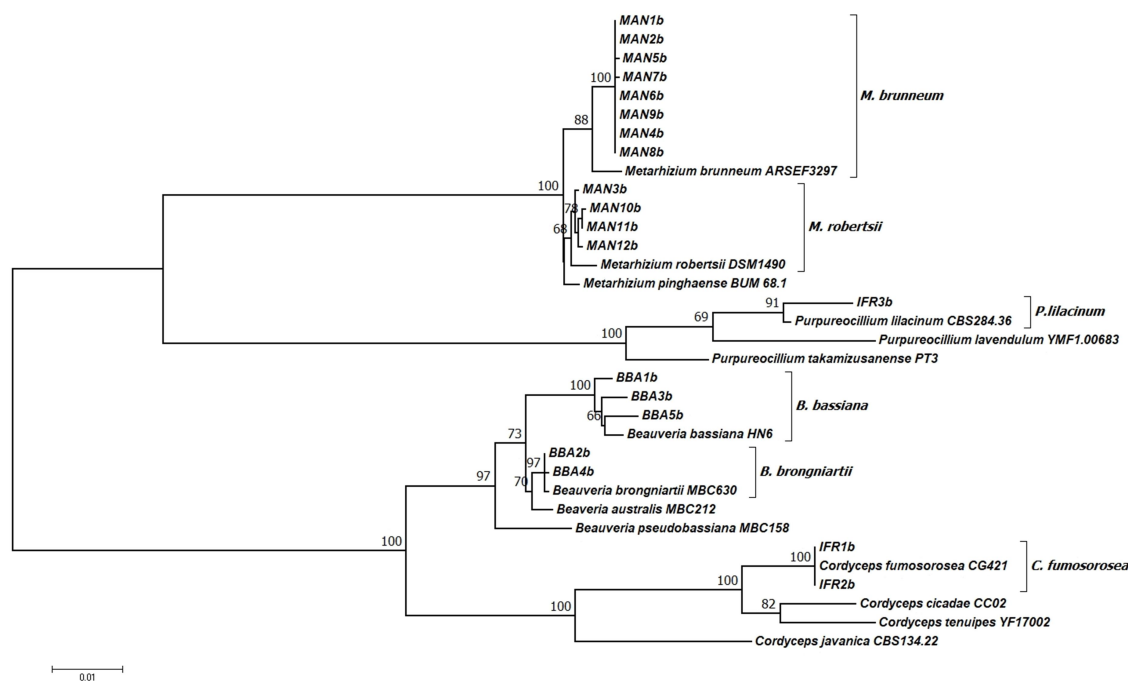


Figure 1. Maximum likelihood tree showing the phylogenetic placement of the isolated strains (for abbreviations see Table 2) and reference fungal species based on three partial loci (nrLSU + ITS + tef1). The ML tree has midpoint rooting and bootstrap values > 65% are shown. Clusters of identified isolates with the reference strains are designated by species names and brackets.

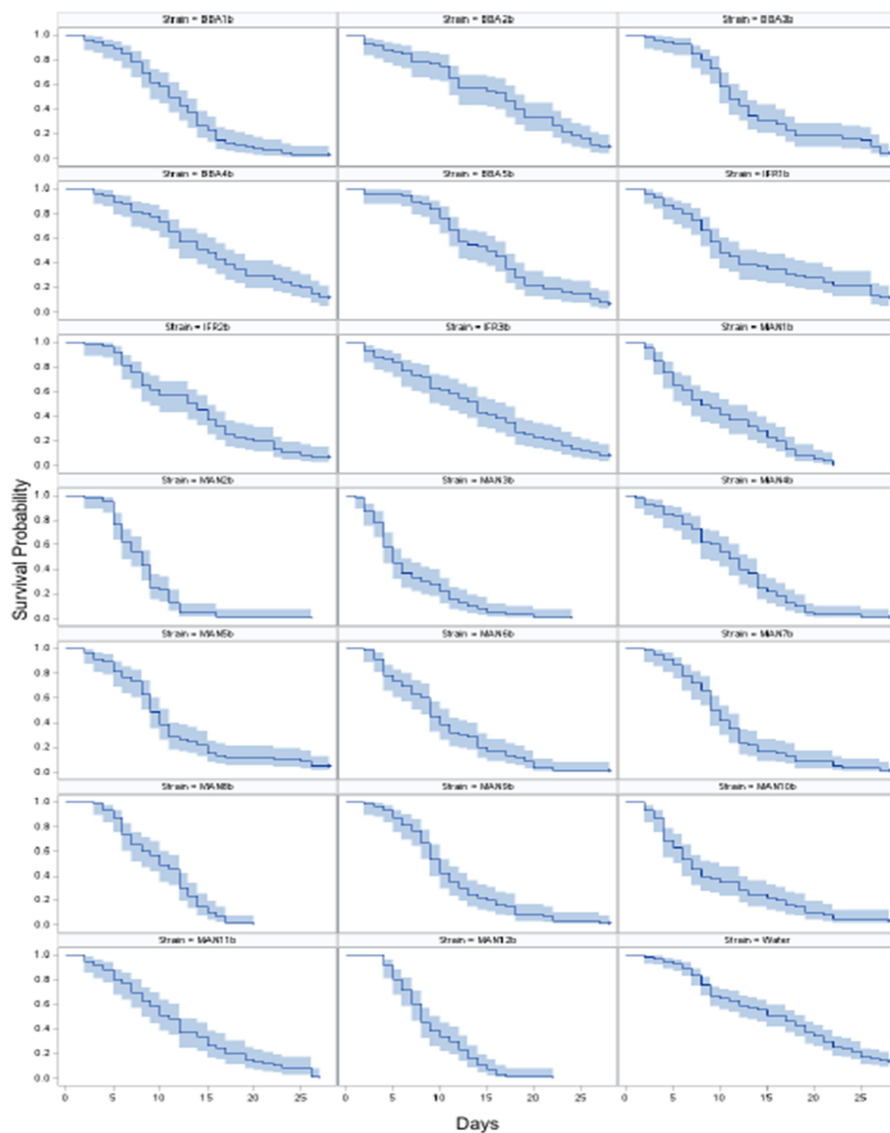


Figure 2. Product-limit survival estimates for *Leptinotarsa decemlineata* adults treated with individual strains of entomopathogenic fungi or with a sterile solution of 0.05% Tween[®] 80 only (control), with 95% confidence limits.

¹ The mean survival time and its standard error were underestimated because the largest observation was censored and the estimation was restricted to the largest event time.

² Significance of the difference to the control group (p values adjusted for multiple comparisons according to Šidák).

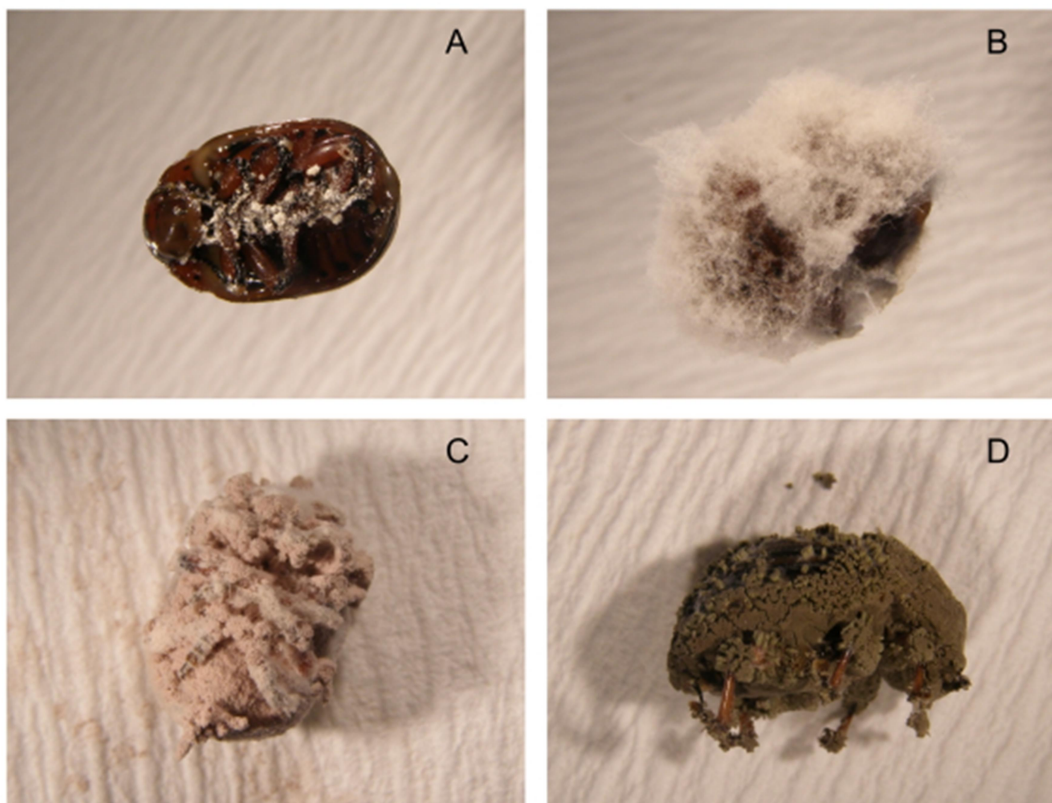


Figure 3. Mycosed cadavers of *Leptinotarsa decemlineata* adults. A, B: *Beauveria bassiana*, C: *Cordyceps fumosorosea*, D: *Metarhizium brunneum*.

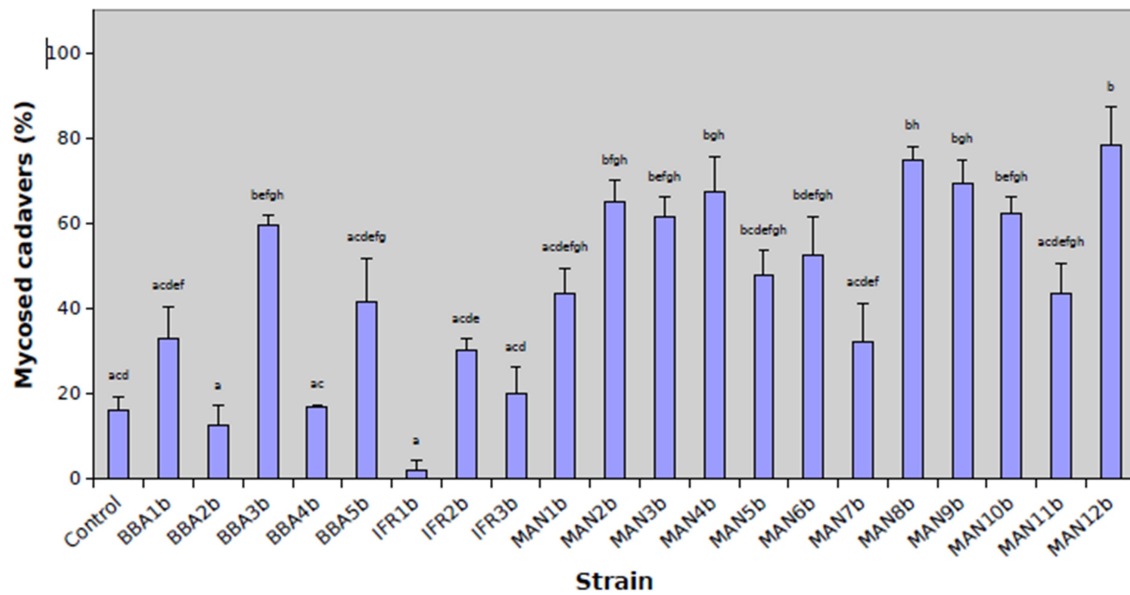


Figure 4. Percentage of mycosed cadavers of *Leptinotarsa decemlineata* adults treated with various strains of entomopathogenic fungi. See Table 4 for the key to the strains. Data presented are means ($\pm$ SE), with three replicates of 20 beetles for each strain. A generalized linear model was fitted and pairwise between treatment differences were tested using the least-square means. Different letters indicate significant differences between columns ($P < 0.05$).

Table 1. Sampling sites in which soil for isolation of EPF was collected.

Label	City/village name	Geographical Coordinates	Crop System
A	Bělčice	49.50702N, 13.89545E	conventional
B	Supíkovice	50.29784N, 17.26655E	conventional
C	Havlíčková Borová	49.62646N, 15.77537E	conventional
D	Žabčice	49.02074N, 16.61598E	conventional
E	Březová lesní Albrechtice	49.80534N, 17.86774E	conventional
F	Malonty	48.69105N, 14.5895E	organic
G	Zálezlice	50.29806N, 14.46234E	conventional
H	Chlebovice	49.66165N, 18.28710E	conventional
I	Chlumín	50.28191N 14.46264E	conventional
J	Hořice	49.08624N, 14.04858E	conventional
K	Hukovice	49.65356N, 18.02546E	conventional
L	Nučničky	50.50541N, 14.21657E	conventional
M	Valečov	49.64343N, 15.49964E	conventional

Table 2. PCR primers and PCR profile.

Forward / Revers		Reference
NL1 (5'-GCATATCAATAAGCGGAGGAAAAG-3') / NL4 (5'-GGTCCGTGTTTCAAGACGG-3')		
LSU		
(28S	Profile	
RNA)	initial denaturation 5 min at 94°C, 25 cycles of denaturation at 94°C 60s, annealing at 50°C 60s, elongation at 72°C 60s, final elongation at 72°C for 5min	modified (O'Donnell 1992, 1993)
ITSf1 (5'-TACACACCGCCCGTCGCTACTA-3') / ITSr1 (5'-CTTTTCCTCCSCTTAYTRATATGC-3)		
ITS	Profile	
	initial denaturation 2 min at 94°C, 35 cycles of denaturation at 94°C 30s, annealing at 50 - 52°C 30s, elongation at 72°C 50s, final elongation at 72°C for 2min	modified (Ji et al. 2003)
EF1-983F (5'-GCYCCYGGHCAYCGTGAYTTYAT-3') / EF1-2218R (5'-ATGACACCRACRGCRACRGTYTG-3)		
EF	Profile	
(tef1)	initial denaturation 2 min at 94°C, 35 cycles of denaturation at 94°C 30s, annealing at 47-52°C 30s, elongation at 72°C 50s, final elongation at 72°C for 2min	Rehner and Buckley 2005

Table 3. List of reference taxa/strains.

Sequences used as outgroups	Strain
Hypocreales, Cordycipitaceae, Beauveria	
<i>Beauveria bassiana</i>	HN6
<i>Beauveria pseudobassiana</i>	MBC158
<i>Beauveria brongniartii</i>	MBC630
<i>Beauveria australis</i>	MBC212
Hypocreales, Cordycipitaceae, Cordyceps	
<i>Cordyceps fumosorosea</i>	CG421
<i>Cordyceps javanica</i>	CBS 134.22
<i>Cordyceps tenuipes</i>	YF17002
<i>Cordyceps cicadae</i>	CC02
Hypocreales, Ophiocordycipitaceae, Purpureocillium	
<i>Purpureocillium lilacinum</i>	CBS 284.36
<i>Purpureocillium lavendulum</i>	YMF1.00683
<i>Purpureocillium takamizusanense</i>	PT3
Hypocreales, Clavicipitaceae, Metarhizium	
<i>Metarhizium brunneum</i>	ARSEF 3297
<i>Metarhizium robertsii</i>	DSM1490
<i>Metarhizium pinghaense</i>	isolate BUM_68.1

Table 4. Strains of entomopathogenic fungi isolated within this study and used in bioassays.

Species	Strain	Sampling site label ¹	Soil sample type	Genbank accession numbers
<i>Beauveria bassiana</i>	BBA1b	A	Potato field	nrLSU: OR758775 ITS: OR758796 tef1: OR767407
	BBA3b	B	Uncultivated boundary	nrLSU: OR758776 ITS: OR758797 tef1: OR767408
	BBA5b	C	Uncultivated boundary	nrLSU: OR758777 ITS: OR758798 tef1: OR767409
<i>Beauveria brongniartii</i>	BBA2b	C	Uncultivated boundary	nrLSU: OR758778 ITS: OR758799 tef1: OR767410
	BBA4b	D	Uncultivated boundary	nrLSU: OR758779 ITS: OR758800 tef1: OR767411
<i>Cordyceps fumosorosea</i>	IFR1b	E	Uncultivated boundary	nrLSU: OR758780 ITS: OR758801 tef1: OR767412
	IFR2b	F	Uncultivated boundary	nrLSU: OR758781 ITS: OR758802 tef1: OR767413
<i>Purpureocillium</i>	IFR3b	G	Uncultivated	nrLSU: OR758782

<i>lilacinum</i>			boundary	ITS: OR758803 tefl: OR767414
<i>Metarhizium brunneum</i>	MAN1b	A	Uncultivated boundary	nrLSU: OR758783 ITS: OR758804 tefl: OR767415
	MAN2b	H	Uncultivated boundary	nrLSU: OR758784 ITS: OR758805 tefl: OR767416
	MAN4b	I	Uncultivated boundary	nrLSU: OR758785 ITS: OR758806 tefl: OR767417
	MAN5b	C	Potato field	nrLSU: OR758786 ITS: OR758807 tefl: OR767418
	MAN6b	C	Potato field	nrLSU: OR758787 ITS: OR758808 tefl: OR767419
	MAN7b	J	Potato field	nrLSU: OR758788 ITS: OR758809 tefl: OR767420
	MAN8b	K	Potato field	nrLSU: OR758789 ITS: OR758810 tefl: OR767421
	MAN9b	L	Potato field	nrLSU: OR758790 ITS: OR758811

				tef1: OR767422
<i>Metarhizium robertsii</i>	MAN3b	B	Uncultivated boundary	nrLSU: OR758791 ITS: OR758812 tef1: OR767423
	MAN10b	M	Uncultivated boundary	nrLSU: OR758792 ITS: OR758813 tef1: OR767424
	MAN11b	M	Uncultivated boundary	nrLSU: OR758793 ITS: OR758814 tef1: OR767425
	MAN12b	G	Potato field	nrLSU: OR758794 ITS: OR758815 tef1: OR767426

¹ See Table 1 for site names.

Table 5. Mean survival time ($\pm$ SE) and median lethal time (LT₅₀) of *Leptinotarsa decemlineata* adults treated with suspensions of strains of entomopathogenic fungi or with a sterile solution of 0.05% Tween[®] 80 only (control).

Treatment	Survival time ¹ (days)	LT ₅₀ (95% CI) (days)	Log-rank test ² (p-values)
BBA1b	11.83 $\pm$ 0.70	11.5 (9.0-13.0)	<0.0001
BBA2b	15.78 $\pm$ 1.04	17.0 (12.0-19.0)	0.8845
BBA3b	13.65 $\pm$ 0.90	11.0 (10.0-13.0)	0.0106
BBA4b	15.77 $\pm$ 0.99	15.0 (12.0-18.0)	0.8805
BBA5b	15.55 $\pm$ 0.87	15.5 (12.0-17.0)	0.4393
IFR1b	13.58 $\pm$ 1.09	10.0 (9.0-14.0)	0.0562
IFR2b	13.53 $\pm$ 0.88	13.5 (9.0-15.0)	0.0121
IFR3b	13.9 $\pm$ 1.02	14.0 (9.0-17.0)	0.0540
MAN1b	9.87 $\pm$ 0.78	8.0 (6.0-11.0)	<0.0001
MAN2b	8.27 $\pm$ 0.48	8.0 (6.0-9.0)	<0.0001
MAN3b	6.90 $\pm$ 0.62	5.0 (4.0-6.0)	<0.0001
MAN4b	11.13 $\pm$ 0.73	11.0 (8.0-13.0)	<0.0001
MAN5b	10.87 $\pm$ 0.82	9.0 (8.0-11.0)	<0.0001
MAN6b	10.02 $\pm$ 0.71	9.0 (7.0-11.0)	<0.0001
MAN7b	10.73 $\pm$ 0.72	9.5 (9.0-11.0)	<0.0001
MAN8b	10.17 $\pm$ 0.53	10.0 (8.0-12.0)	<0.0001
MAN9b	11.07 $\pm$ 0.71	10.0 (9.0-11.0)	<0.0001
MAN10b	9.68 $\pm$ 0.92	7.0 (5.0-10.0)	<0.0001
MAN11b	11.98 $\pm$ 0.89	11.0 (8.0-12.0)	<0.0001
MAN12b	9.18 $\pm$ 0.51	8.0 (7.0-10.0)	<0.0001
Control	16.21 $\pm$ 0.74	17.0 (12.0-19.0)	-